

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA4303

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COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS:

Total Marks: 30

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">
  <input type="text" name="name">
  <input type="email" name="email">
  <button>Submit</button>
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly.

Questions:

1. Identify appropriate **HTML5 semantic elements** missing in the structure.
2. Modify the structure to make it **standards-compliant**.
3. Interpret the **HTTP request generated** during form submission.
4. Analyze the **HTTP response cycle** from server to browser.
5. Suggest improvements for **usability and accessibility**.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>
  <head>
    <title>Company</title>
  </head>
  <body>
    <div>
      <h1>Welcome
      <p>About us
    </div>
  </body>
</html>
```

Questions:

1. Identify **improper nesting and missing closing tags**.
2. Analyze how different browsers may interpret this structure.
3. Identify **missing semantic elements** (e.g., header, section).
4. Provide a **corrected W3C-compliant HTML structure**.
5. Suggest techniques to ensure **cross-browser compatibility**.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">
  <input type="text" name="username">
  <input type="password" name="password">
  <button type="button">Login</button>
</form>
```

Questions:

1. Identify issues in the **form configuration**.
2. Analyze why the **HTTP request is not triggered**.
3. Interpret the role of **GET method in this scenario**.
4. Propose a **corrected implementation**.
5. Suggest improvements for **secure data transmission**.

Code of Conduct:

I G.MOKSHAGNA(192411160)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G.MOKSHAGNA

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Online Symposium Portal

1. Identify appropriate HTML5 semantic elements missing in the structure.

- `<header>`
- `<main>`
- `<section>`
- `<footer>`

2. Modify the structure to make it standards-compliant.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Symposium Registration</title>
```

```
</head>
```

```
<body>
```

```
<header>
```

```
<h1>Online Symposium Registration</h1>
```

```
</header>
```

```
<main>
```

```
<section>
```

```
<form action="/register" method="POST">
```

```
<label for="name">Name:</label>
```

```
<input type="text" id="name" name="name" required>
```

```
<label for="email">Email:</label>
```

```
<input type="email" id="email" name="email" required>
```

```
<button type="submit">Submit</button>
```

```
</form>
```

```
</section>
```

```
</main>
```

```
<footer>
```

```
<p>&copy; 2026 College Symposium</p>
```

```
</footer>
```

```
</body>
```

```
</html>
```

3. Interpret the HTTP request generated during form submission.

- The browser sends an **HTTP POST** request to **/register**.
- The form data (name and email) is sent in the request body.

4. Analyze the HTTP response cycle from server to browser.

- Browser sends the request.
- Server processes the registration.
- Server returns an HTTP response (e.g., **200 OK** or **201 Created**).
- Browser displays the confirmation page.

5. Suggest improvements for usability and accessibility.

- Add `<label>` elements for all inputs.
- Use the `required` attribute.
- Provide error messages for invalid input.
- Ensure keyboard accessibility.
- Add proper headings and semantic HTML elements.

The screenshot displays a web application titled "Online Symposium Registration Portal". It features a navigation bar with links for Home, Events, Register, and Contact. The main content area contains a "Student Registration" form with two input fields: "Student Name" (containing "moksha") and "Email Address" (containing "192411160.simats@saveetha.com"). A "Register" button is positioned below the email field. The footer of the page includes the copyright notice "© 2026 College Symposium | All Rights Reserved".

2. Cross-Browser Compatibility Issue

1. Identify improper nesting and missing closing tags.

- Missing `</h1>`
- Missing `</p>`

2. Analyze how different browsers may interpret this structure.

- Browsers automatically close missing tags.
- Different browsers may render the page slightly differently.

3. Identify missing semantic elements.

- `<header>`
- `<main>`
- `<section>`

4. Provide a corrected W3C-compliant HTML structure.

```
<!DOCTYPE html>
<html>
<head>
<title>Company</title>
</head>
<body>

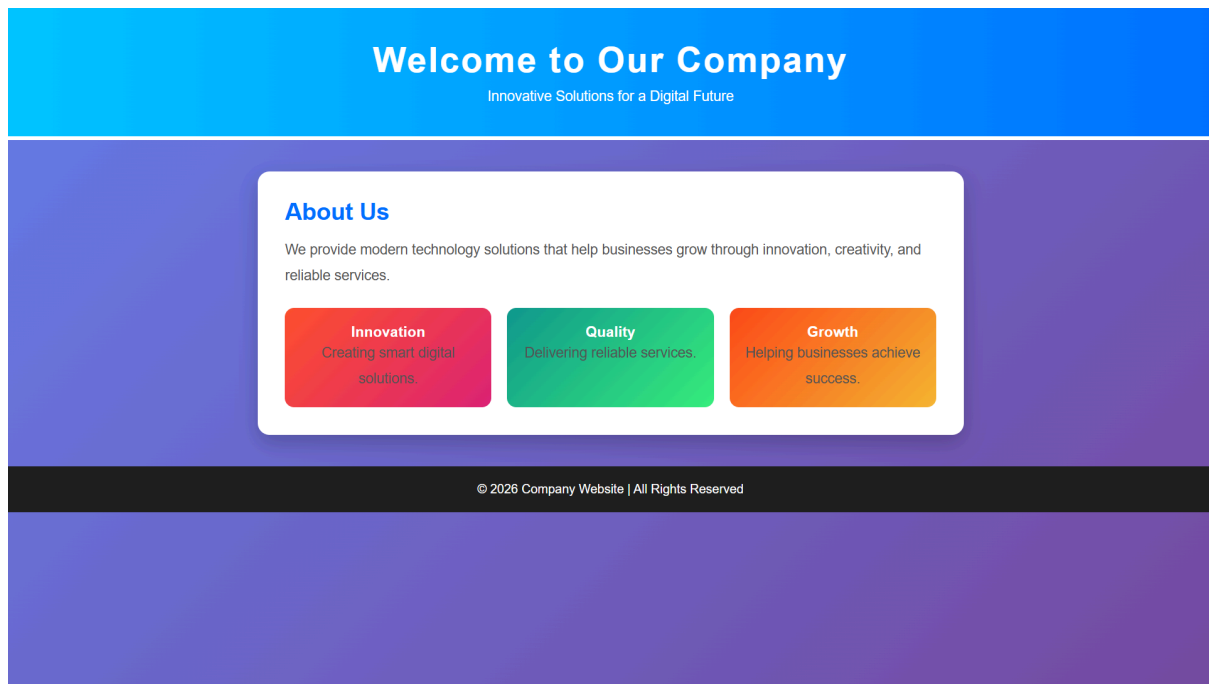
<header>
<h1>Welcome</h1>
</header>

<main>
<section>
<p>About us</p>
</section>
</main>

</body>
</html>
```

5. Suggest techniques to ensure cross-browser compatibility.

- Use valid HTML5 and CSS.
- Test on multiple browsers.
- Use CSS reset/normalize.
- Validate code using the W3C Validator.
- Avoid deprecated HTML elements.



3. Form Submission Failure

1. Identify issues in the form configuration.

- `method="GET"` is not suitable for passwords.
- `<button type="button">` does not submit the form.

2. Analyze why the HTTP request is not triggered.

- The button type is **button**, so clicking it does not submit the form.

3. Interpret the role of GET method in this scenario.

- GET sends data in the URL.
- It is not secure for login credentials because the password becomes visible in the URL.

4. Propose a corrected implementation.

```
<form action="/submit" method="POST">
```

```
<label for="username">Username:</label>
```

```
<input type="text" id="username" name="username" required>
```

```
<label for="password">Password:</label>
```

```
<input type="password" id="password" name="password" required>
```

```
<button type="submit">Login</button>  
</form>
```

5. Suggest improvements for secure data transmission.

- Use the **POST** method.
- Enable **HTTPS**.
- Encrypt passwords before storing them.
- Validate user input.
- Use secure session management after login.